

David McClung

Turning manual processes into software people use | Software, digital transformation and AI automation

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PROFILE

I have just completed an MSc in Artificial Intelligence in Business at Queen's University Belfast, and am due to graduate in December 2026. My summer internship at Tetra Tech was my first software job. The work I enjoy most is sitting down with a team, learning how they do a job today and building software that takes the repetitive parts off their hands, so they have more time for the work that needs them. I'm now looking for my first role in software, digital transformation or AI automation.

KEY SKILLS

- AI-Assisted Development (Claude Code)
- Backend Development (REST, Laravel, Django, Python)
- Frontend Development (Vue 3, TypeScript, PrimeVue)
- Structured Data & JSON Schemas
- SQL & Relational Databases
- Automated Testing (Pest, PHPUnit)
- Git & CI/CD (Azure Pipelines)
- Agile Delivery (Scrum)
- Process Mapping & User Adoption
- Stakeholder Communication & Reporting

EXPERIENCE

Data and AI Intern | Tetra Tech, Summer 2026

- Used AI-assisted coding tools (including Claude Code) to help build and maintain Bat Analytics Pro, a Laravel REST API and Vue.js SPA processing ecology survey and weather data for the internal ecology team, automating the charts and tables the team had been building by hand.
- Helped to design a structured, schema-versioned JSON data package that consolidates processed survey data for an external AI summarisation service, including a signed, short-lived token for secure service-to-service authentication.
- Helped to specify validation gating logic so AI summary generation is blocked until required data entry is confirmed complete, an early human-in-the-loop control against acting on incomplete data.
- Added unit, feature and endpoint tests (Pest/PHPUnit) to a CI pipeline (Azure Pipelines).
- Worked in a Scrum team; mapped the ecology team's manual reporting process and scoped requirements with them, then demoed the app and helped the team adopt it, refining it through regular check-ins.

Treasurer | Newtownabbey Independent Christian School, March 2025 – Present

- Manage financial reporting and oversight of a c.£100k annual budget on a voluntary basis, demonstrating accountability, integrity and attention to detail.
- Present monthly financial reports to the management committee, communicating budget performance clearly to a non-technical audience.
- Built Ledger (Python, Django) in August 2026, after more than a year of manual Excel bookkeeping; entered the previous financial year into it, and the first live month took under 30 minutes, down from several evenings.

Additional Experience | Premier Sound Solutions / PDG Electrical / FJM Logistics

- Media Technician / Consultant – designed and implemented audio-visual systems for clients, communicating technical requirements clearly and training 50+ end users.
- Electrician's Assistant – delivered 100+ installation and maintenance projects, troubleshooting on-site issues within a team of eight.
- Delivery Associate – delivered 300+ packages daily, consistently meeting deadlines and activity targets.

PROJECTS

Bat Analytics Pro | Tetra Tech internship

- Laravel API and Vue app that turns ecology survey and weather data into report charts and tables, and sends that data to an external AI service for report summaries.

Ledger | Volunteer treasurer, in use

- Python and Django app that took over the repetitive work in the school's Excel-based accounts; holds the previous financial year, including 12 monthly reports and the annual report.

Premier Sound Solutions website | Freelance, in development

- Vue 3 and TypeScript site replacing the company's old WordPress site, built to win new clients and show their installations.

EDUCATION

In progress: Udemy courses: Full-Stack Web Development Bootcamp; 100 Days of Code: The Complete Python Pro Bootcamp

2025 – 2026: MSc: AI in Business, Queen's University Belfast (graduating December 2026)

2021 – 2024: Bachelor of Arts: Broadcast Production, Queen's University Belfast

2020 – 2021: 3 A Levels: A in Digital Technology, A in Geography, B in Business Studies; 1 AS Level: B in Design & Technology